# Temporal dynamics of argument-verb computations in sentence processing: New evidence from Turkish

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## Introduction

**Prediction in language comprehension:** Comprehenders use contextual information to anticipate upcoming linguistic input and generate predictions in many aspects of sentence processing (Altmann & Mirković, 2009; Kuperberg & Jaeger, 2016; Pickering & Gambi, 2018).

- N400s have been used as an index of lexical-semantic predictions (e.g., Kutas & Hillyard, 1980; Lau et al., 2008). N400 amplitude is typically modulated by offline cloze probability:
  - **Higher predictability → smaller N400 amplitude**

**Verb prediction:** Results from EEG studies appear to show that **N400 may be insensitive to argument role information** in sentences with non-canonical argument roles although offline cloze norming shows stronger expectation for a role-consistent verb in canonical contexts (Chow et al., 2016; 2018; Liao et al., 2022; Nakamura et al., 2024; Stone & Rabovsky, 2025):

- **Canonical:** They know which thief the cop ***arrested***.
- **Role reversal:** They know which cop the thief ***arrested***.

Although ***arrested*** is used more in the canonical condition in offline cloze norming tasks, EEG studies do not find an immediate N400 facilitation for the canonical verb

**Why might verb prediction be different?**

- Liao et al. (2022) proposed that verb prediction unfolds in three distinct stages:
  1. **Bag of Words:** Lexical associations among any preverbal words
  2. **Bag of Arguments:** Identification of arguments vs. non-arguments, narrowing the set of candidate verbs
  3. **Argument Roles:** Assignment of thematic roles (*who did what to whom*), allowing access to event knowledge and role-consistent verbs
- Previous studies have tested different stages of this process in different languages and structures with different participants (Chow et al., 2016, 2018; Liao et al., 2022)
- **The present study** examines the precise time-course of argumenthood and argument role information in Turkish, using the genitive-possessive construction to test when and how these cues are used during verb prediction with the same participants

## Experiment 1: Argumenthood

| Target sentences (Argumenthood) | | | Average cloze probability |
|---|---|---|---|
| **1a** | Verb-related Non-argument | Dün **patron** [**sekreter-in** **terzi-si**]-ni **kov-du** galiba.<br>yesterday **boss** [**secretary-GEN** **tailor-POSS**]-ACC **fire-PAST** apparently<br>'Apparently, the boss fired the secretary's tailor yesterday.' | 1.75 |
| **1b** | Verb-unrelated Non-argument | Dün **patron** [**dişçi-nin** **terzi-si**]-ni **kov-du** galiba.<br>yesterday **boss** [**dentist-GEN** **tailor-POSS**]-ACC **fire-PAST** apparently<br>'Apparently, the boss fired the dentist's tailor yesterday.' | 2 |

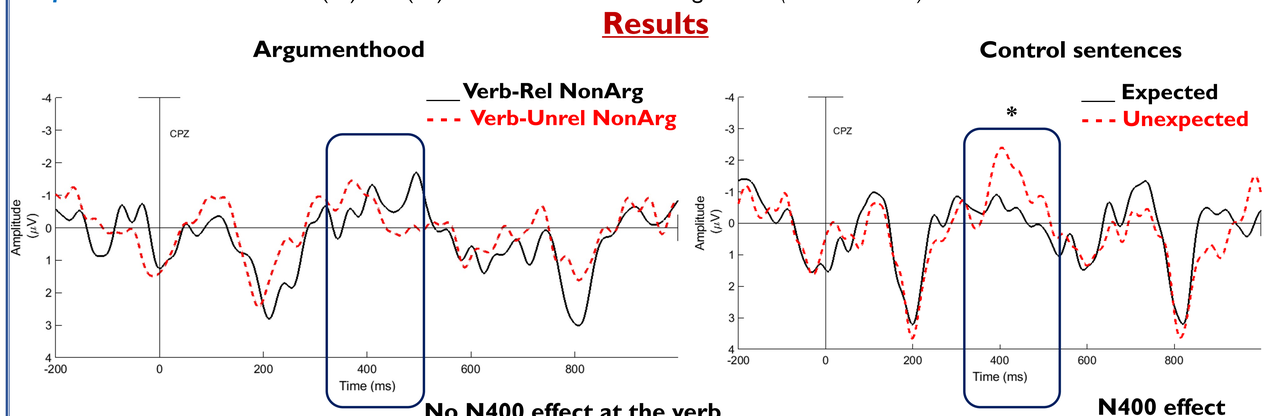
- Same structure + same verb (***fired***) + same arguments (***boss*** and ***tailor***) in (1a) and (1b)
- Genitive-marked nouns are **NOT** arguments: ***secretary* in (1a) & *dentist* in (1b)**
  - Although genitive-marked nouns can serve as subjects of nominalized embedded clauses in Turkish complex sentences, norming data showed that participants completed sentence fragments with a main verb rather than an embedded-clause structure
- ***secretary* is related to *fired*** in (1a) & ***dentist* is unrelated to *fired*** in (1b)

| Control sentences (Lexical-semantic predictions outside of verb prediction domain) | | | Average cloze probability |
|---|---|---|---|
| **2a** | Expected Condition | Yağmurlu gün-ler-de yanımda daima **şemsiye** taşı-r-ım.<br>rainy day-PL-LOC with me always **umbrella** carry-AOR-1stSG<br>'I always carry an **umbrella** with me on rainy days.' | 42.47 |
| **2b** | Unexpected Condition | Yağmurlu gün-ler-de yanımda daima **pencere** taşı-r-ım.<br>rainy day-PL-LOC with me always **window** carry-AOR-1stSG<br>'I always carry a **window** with me on rainy days.' | 0 |

### Predictions

- **Control sentences: Smaller N400s** for the Expected Condition (2a) than Unexpected Condition (2b)
- **Bag of Words predictions:** If only the Bag of-Words mechanism has gone into the computation, we predict **smaller N400s** for the verb in (1a) as lexical associations between ***boss*** and ***secretary,*** and ***fired*** in the Verb-related Non-argument condition (1a) should facilitate the prediction of ***fired*** more than in Verb-unrelated Non-argument condition (1b)
- **Bag of Arguments predictions:** If the Bag-of-Arguments mechanism has gone into the computation, **N400s** at the verb ***fired*** should **NOT** differ in (1a) and (1b) because the structural arguments (***boss & tailor***) are identical.

### Results




**No N400 effect at the verb** | **N400 effect**

- The presence of an additional noun that was related to the verb (***fired***) but not a structural argument of it (***e.g., secretary***) did not facilitate verb prediction
- Bayes factors (BF) analysis also showed substantial evidence for the null hypothesis over the alternative hypothesis (BF = 0.28)

- Smaller N400s in the Expected Condition (2a) than Unexpected Condition (2b)
- Evidence that the same participants did show N400 effect in contexts outside of the Argumenthood conditions

**Evidence for the use of the 'Bag of Arguments' mechanism at 600 ms argument-verb interval**

## Methodology

**Participants:** 17 adult L1-Turkish speakers living in the USA (data collection ongoing)

- 11 females; 6 males; Age range: 19-48, *M* = 33.71, *SD* = 6.49
- High vocabulary scores in Turkish (Turkish Vocabulary Test: *M* = 54.53 out of 60)
- High reading experience scores (Author Recognition Test in Turkish: *M* = 68.07 out of 80)

**Procedure:**

- Items were normed for predictability in three separate cloze norming experiments (Total N = 165)
- Each participant read 300 sentences while their brain activity was recorded using EEG:
  - 60 Experiment 1 target items; 120 Experiment 2 target items; 60 control items; 60 additional fillers (1:1 plausibility ratio)
- Experiments 1 and 2 were completed in a single session by the same participants
- Sentences were presented word-by-word with an SOA of 600 ms (400 ms per word + 200 ms pause; Chow et al., 2018)
- EEG data were analyzed based on mean amplitudes from nine centro-parietal electrodes (Lee & Phillips, 2025; Liao et al., 2022) for the following time windows: N400: 300-500 ms; P600: 600-800 ms
- **Comprehension task:** After 20% of the items, participants saw a sentence, and they were asked whether it was a correct paraphrase of the sentence they had just read (Liao et al., 2022)
  - Mean accuracy: 90% (*SD* = 3.32, Range: 81.7% - 95%)

## Experiment 2: Argument role information

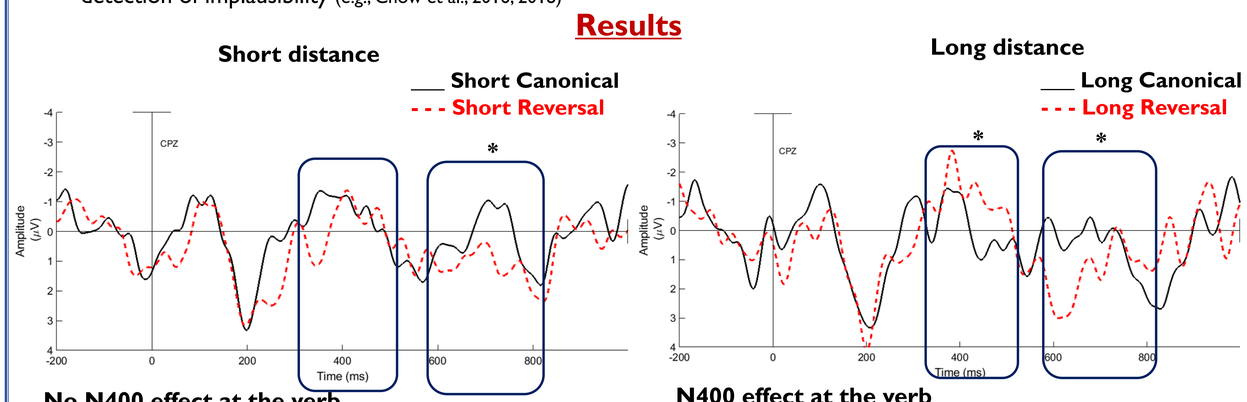
| Target sentences (Argument role information) | | | Average cloze probability |
|---|---|---|---|
| **3a** | Short, Canonical | Dün **patron** Ayşe-nin **sekreter-i**-ni **kov-du** galiba.<br>yesterday **boss** Ayşe-GEN **secretary-POSS**-ACC **fire-PAST** apparently<br>'Apparently, the boss fired Ayşe's secretary yesterday.' | 40.92 |
| **3b** | Short, Reversal | Dün **sekreter** Ayşe-nin **patron-u**-nu **kov-du** galiba.<br>yesterday **secretary** Ayşe-GEN **boss-POSS**-ACC **fire-PAST** apparently<br>'Apparently, the secretary fired Ayşe's boss yesterday.' | 0.45 |
| **3c** | Long, Canonical | **Patron** Ayşe-nin **sekreter-i**-ni **dün** **kov-du** galiba.<br>**boss** Ayşe-GEN **secretary-POSS**-ACC **yesterday** **fire-PAST** apparently<br>'Apparently, the boss fired Ayşe's secretary yesterday.' | 40.90 |
| **3d** | Long, Reversal | **Sekreter** Ayşe-nin **patron-u**-nu **dün** **kov-du** galiba.<br>**secretary** Ayşe-GEN **boss-POSS**-ACC **yesterday** **fire-PAST** apparently<br>'Apparently, the secretary fired Ayşe's boss yesterday.' | 0.43 |

- **Short distance (3a&b):** The verb immediately follows the second argument (600 ms argument-verb interval)
- **Long distance (3c&d):** Inserting a neutral time adverbial (e.g., ***yesterday***) increased the argument-verb interval to 1200 ms.

### Predictions

- **N400 predictions:** If argument-role information only impacts prediction with additional time (e.g., Chow et al., 2018):
  - **Short distance (3a&b):** No N400 difference at the verb in (3a) and (3b)
  - **Long distance (3c&d):** The emergence of N400 effect (Smaller N400s at the verb in (3c) than (3d))
- **P600 predictions:** Larger P600s at the verb for Reversal (3b, d) than Canonical conditions (3a, c)
  - Some EEG studies also report **larger P600s** for role-reversal than canonical sentences, which they suggest reflects the detection of implausibility (e.g., Chow et al., 2016; 2018)

### Results




**No N400 effect at the verb**
**P600 effect at the verb**

**N400 effect at the verb**
**P600 effect at the verb**

**Evidence for the use of argument role information only at 1200 ms argument-verb interval**

## General discussion

- We examined **the time precise course of verb prediction in Turkish, testing the same participants and using the same structure within the same study**
- The results show that **comprehenders can predict verbs in Turkish**, and that **different sources of information contribute to prediction at different points in time**:
- **Experiment 1: Evidence for the use of the 'Bag of Arguments' mechanism at 600 ms argument-verb interval**
- **Experiment 2: Evidence for the use of argument role information only at 1200 ms argument-verb interval**
  - Our results are **consistent with the staged model of verb prediction** proposed by Liao et al. (2022)
- These findings help to underscore the value of precisely examining not only which information sources can be used but also the precise temporal dynamics under which they can be utilized in particular contexts and under different task demands (e.g., Lee & Phillips, 2025; Nakamura et al., 2024)

**References:** Altmann & Mirković (2009). *Lang, Cogn and Neurosci, 33*, 583-609. Chow et al. (2016). *Lang, Cogn and Neurosci, 33*(7), 1-26. Chow et al. (2018). *Lang, Cogn and Neurosci, 31*(5), 577-596. Kuperberg & Jaeger (2016). *Lang, Cogn and Neurosci, 31*, 32-59. Kutas & Hillyard (1980). *Science, 207*, 203-205. Kutas & Hillyard (1980). *Nature, 307*, 161-163. Lau et al. (2008). *Nature Rev. Neurosci, 9*, 920-933. Lee & Phillips (2025). *Cognition, 264,106255.* Liao et al. (2022). *JML, 126*, 104350. Nakamura et al. (2024). *Cogn Sci, 48*(12), e70023. Pickering & Gambi (2018). *Psych. Bull., 144*, 1002. Stone & Rabovsky (2025). *J. of Cog. Sci*, 1-27.

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